A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 10

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 4 points below. **1 Gravimetry:** gravimeters detect variations in gravity caused by differences in rock density / mass (metal ores usually denser than sedimentary rock) **1** Magnetometry **1 Seismic survey:** sound / shock waves from controlled explosions or a seismic vibrator – echoes recorded by geophones give the depth, density and shape of rock strata **1** Resistivity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Trial drilling is the most expensive technique per site **1** Remote sensing (satellite, aircraft, ground surveys) covers large areas at low cost per unit area and identifies promising sites **1** Drilling is then targeted where deposits are likely – it is the only method that produces actual rock samples for chemical analysis to confirm composition and purity (proven reserves) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **remote sensing** → collecting data from aircraft or satellites	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Different minerals emit / reflect infrared at different wavelengths **1** Multispectral sensors detect these wavelengths over the whole surface (eg Landsat 8 every 16 days) giving information on surface mineral composition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2
01.4	1 mark for the point below. **1** Results are available immediately in the field, saving time waiting for laboratory analysis **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 137; AQA spec 3.2.3.4.	
02.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Steam washing (hot water / steam jets kill organisms in the sediment)	1
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Chemicals cause the oil molecules to join together / solidify **1** Producing more solid material that can be collected / lifted more easily than liquid oil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Air contains about 210 cm^3 of oxygen per litre whereas water at 20 °C holds a maximum of about 6 cm^3 per litre (less than 3% of the amount in air) **1** So a small amount of oxygen-consuming activity (decomposition, respiration) or warming can remove most of the oxygen – many aquatic organisms are near the lower end of their range of tolerance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deoxygenation** → loss of dissolved oxygen from the water • **range of tolerance** → the range of conditions in which a species can survive	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 254-262; AQA spec 3.4.3.2.6-3.4.3.2.7.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Visible light passes through the atmosphere easily and is absorbed by the Earth's surface, which warms and emits infrared **1** Greenhouse gases (naturally water vapour and CO2) absorb infrared better than the average for the atmosphere, converting it to heat and re-emitting some back to the surface **1** This keeps the surface about 33 °C warmer than it would otherwise be **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem	3
03.2	**1 mark each** 1 mark for each of the 3 points below. **1** CO2 is emitted in far larger quantities (every combustion process, deforestation) so its total effect is greatest **1** CO2 has a long residence time so it accumulates – methane has a short residence time (12 years) so it is removed faster **1** CFC emissions are now very small following the Montreal Protocol **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **anthropogenic** → human-caused / caused by human activity • **CFCs** → CFCs / halogenated organic compounds releasing chlorine • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	3

Do not write outside the box

Question	Marking guidance	Marks
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Methane has a much higher global warming potential (28x) so each tonne removed has a bigger effect **1** Its short residence time (about 12 years) means atmospheric concentrations fall quickly once emissions are cut **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **global warming potential** → warming effect compared with the same mass of carbon dioxide • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks.	2
03.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Collect landfill / anaerobic digester gas and use as fuel **1** Changes in cattle diet / fewer cattle / less meat consumption **1** Draining rice paddies periodically **1** Anaerobic digestion of manure with gas capture **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-90; AQA spec 3.2.1.2.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** The water was recharged during or before the last ice age (10 000-20 000 years ago) when the climate was wetter **1** Current recharge is little or none so any abstraction reduces the stored volume **1** The large volume means shortages take a long time to appear, but eventually the supply runs out – ancient water is often saline, causing soil salinisation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abstraction** → water taken out of a river, lake or aquifer • **aquifer** → a permeable rock that stores groundwater / underground water store • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** Two satellites whose orbits are affected by the force of gravity **1** The mass of water in aquifers below changes gravity, so changes in the orbit / distance between the satellites indicate changes in groundwater mass **1 Advantage:** the whole Earth is surveyed every 30 days, whereas borehole water-table measurements are slow and hard to coordinate over large areas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store	3
04.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Low-pressure irrigation pipes to reduce leaks **1** Growing crops that need less water than wheat / maize **1** Charging consumers by volume used (metering) **1** Inter-basin transfer of water 450 km from the Yangtze **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
04.4	1 mark for the point below. **1** Trees intercept rainfall and transpire large volumes, reducing soil moisture and the water percolating down to the aquifer **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 119-121; AQA spec 3.2.2.1-3.2.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Temperature:** most chemical / biological reactions are faster at higher temperature so degradable pollutants (eg sewage) break down faster – but faster decomposition may increase deoxygenation **2 Light:** provides the activation energy for chemical reactions – photodegradation of pesticides (eg neonicotinoids break down fastest in sunlight) – light drives photochemical smog reactions producing secondary pollutants **2 Oxygen:** needed for aerobic bacterial decomposition of organic wastes / sewage – oxidation of sulfide ores produces sulfur dioxide / acid mine drainage – oxidation state affects the solubility of metals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **deoxygenation** → loss of dissolved oxygen from the water • **photochemical smog** → ground-level ozone and PANs formed by sunlight acting on oxides of nitrogen and VOCs **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • ground-level ozone described as protecting us from UV - that is stratospheric ozone; at ground level it is a pollutant	6
05.2	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Pollutant ions adsorb onto (the surface of) clay particles / organic matter in soil or sediment **1** So they are immobilised and cannot be absorbed by organisms / leached into water – radiation is absorbed in the ground **1** But they may be released later if conditions change (eg storms disturbing sediment, low pH) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorb** → nutrient ions sticking to the surfaces of clay and humus • **leached** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Many heavy metals (eg lead, aluminium) are more soluble at low pH **1** So they are mobilised / leached from soil or sediments and can be absorbed by organisms / enter water bodies **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 240-242, 255; AQA spec 3.4.2.2.	
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** A waste is assumed to cause pollution until research shows it is unlikely to do so **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	1
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** If the cost of preventing damage is less than the (penalty for) damage there is a financial incentive to prevent pollution **1** If polluters can escape their responsibilities / penalties are low, there is little incentive to comply **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
06.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Marine discharges where currents dilute and disperse emissions **1** Emissions downwind of urban areas **1** Not discharging waste onto permeable rock above an aquifer **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **permeable** → how easily water passes through / lets water through • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-244; AQA spec 3.4.3.1.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 110 x 0.67 x 1.35 x (0.50 x 0.90) x 1.0 **1** = 44.8 t ha^{-1} yr^{-1} (accept 44.7-44.8) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (t ha^{-1} yr^{-1}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
07.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 110 x 0.67 x 1.35 x (0.35 x 0.60) x 0.25 **1** = 5.2 t ha^{-1} yr^{-1} **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (t ha^{-1} yr^{-1}) is already printed on the answer line. • **contour ploughing** → ploughing along the slope rather than up and down it **Do not credit** • A bare wrong answer with no working.	2
07.3	**1 mark each** 1 mark for each of the 3 points below. **1 Beans / up-and-down ploughing:** severe class (over 30) and about 45 times the formation rate - clearly unsustainable, soil depth will fall **1** Wheat / mulch / strip cropping: moderate class, reduced by about 88% but still 5 times the formation rate so not fully sustainable **1** Further measures (long-term crops, terracing, windbreaks, more organic matter) or grazing land (C = 0.02) needed to bring loss below 1 t ha^{-1} yr^{-1} **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainability** → meeting present needs without stopping future generations meeting theirs • **terracing** → cutting the slope into level steps **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3

Do not write outside the box

Question	Marking guidance	Marks
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Factors are averages / arbitrary values for typical conditions – rainfall erosivity varies between years / storms **1** Does not include gully erosion, landslides, wind erosion or local variation in soil – based on data from particular regions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.5	1 mark for the point below. **1** C (crop / tillage) or P (erosion control practice) - R, K and LS are fixed by climate, soil and topography **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 167-169; AQA spec 3.2.5.2.	
08.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 15 - (-18) = 33 °C **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (°C) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working. • 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem	1
08.2	**1 mark each** 1 mark for each of the 4 points below. **1** Visible light passes easily through the atmosphere and is absorbed by the surface, which warms **1** The warm surface emits (long-wave / far) infrared radiation **1** Greenhouse gases (water vapour, CO_2, methane) absorb infrared better than the average for the atmosphere and convert it to heat **1** The warm atmosphere emits infrared which is absorbed by the surface, and it reduces heat loss from land and oceans by conduction **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **long-wave** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem	4

Do not write outside the box

Question	Marking guidance	Marks
08.3	1 mark for the point below. **1** Any gas that is better at absorbing infrared radiation than the average for the atmosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	1
08.4	**1 mark each** 1 mark for each of the 2 points below. **1 Arriving:** mainly ultraviolet, visible light and near infrared (short wavelength) – leaving: mainly long-wavelength (far) infrared **1** Gases that are transparent to the incoming short wavelengths can absorb the outgoing long wavelengths, so changing their concentration changes how much energy is retained **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2
08.5	**1 mark each** 1 mark for each of the 2 points below. **1** The energy and the processes it drives control climate, ocean currents and the hydrological cycle **1** These determine the abiotic conditions (temperature, rainfall) that each species is adapted to, so a change alters where species can survive **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87-89; AQA spec 3.2.1.1.	
09.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (15.0 - 2.2) / 15.0 x 100 **1** = 85.3% (accept 85%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
09.2	**1 mark each** 1 mark for each of the 3 points below. **1** Added as an anti-knock agent to smooth combustion and reduce engine wear **1** Lead particulates were released in exhaust gases into the air of cities and inhaled / deposited on soil and food **1** Lead is a neurotoxin that bioaccumulates – chronic exposure damages the nervous system / brain development in children **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **particulates** → fine solid particles / smoke and dust	3
09.3	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Replacement of lead water pipes / lead solder with copper / tin-based solder – phosphoric acid dosing of water supplies forming insoluble lead phosphate **1** Phasing out of lead-based paint / safer paint removal **1** Lead-free electrical solder **1** Restrictions on lead fishing weights / shotgun pellets **1** Industrial emission controls **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** Reacts with the lead in old pipes to form an insoluble layer of lead phosphate on the inside of the pipe **1** This prevents lead dissolving into the water **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.5	1 mark for the point below. **1** Swans / ducks swallow them (mistaking them for snails / grit) and the lead is absorbed, causing lead poisoning **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-276; AQA spec 3.4.3.2.11.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**1 mark each** 1 mark for each of the 2 points below. **1** t-test (or Mann-Whitney U if not normally distributed) **1** Comparing two means of counted / measured data from two independent samples **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **t-test** → a test for a difference between the means of two normally distributed sets • **Mann-Whitney** → a test for a difference between two sets of unpaired, non-normal data **Do not credit** • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	2
10.2	**1 mark each** 1 mark for each of the 2 points below. **1** The difference is statistically significant (less than 1% probability that it arose by chance) **1** The null hypothesis (no difference between fields) is rejected – the organic field has significantly more earthworms **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **statistically significant** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance • **null hypothesis** → a statement that there is no significant difference or correlation **Do not credit** • 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it	2
10.3	**1 mark each** 1 mark for each of the 2 points below. **1** A significant difference does not show a causal relationship **1** Other variables (soil type, moisture, pH, cultivation history) may differ between the two fields and cause the difference **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **causal** → one variable actually causing the change in the other **Do not credit** • 'organic farming uses no pesticides' - a restricted list of pesticides is permitted	2

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Scientific methodology:** identifying the topic, reviewing existing knowledge, formulating a testable hypothesis and null hypothesis, designing the study, analysing and drawing conclusions, planning further research • **Sample location:** random (grid coordinates / numbered sites) or systematic (transects, fixed intervals) to avoid bias – interval chosen so variation is detected without excess data • **Sample timing:** diurnal, seasonal, weather-related and long-term changes – sampling on different occasions – interval matched to rate of change • **Sample size and number:** larger and more samples for non-homogeneous variables – running mean and standard deviation show when the mean is reliable – preliminary study to decide • Standardisation and control of variables – accuracy, precision, reliability, representativeness – anomalies and repeats – calibration of equipment – appropriate techniques and risk management • **Statistical analysis:** choosing the test (Spearman, chi-squared, t-test, Mann-Whitney), significance and p-values, error bars, correlation vs causation • **Evaluation:** trade-offs between cost, time, disturbance and reliability – limitations of sub-sampling – justified conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9

Do not write outside the box

Question	Marking guidance	Marks
	• **valid** → actually measures what it is meant to measure • **null hypothesis** → a statement that there is no significant difference or correlation • **random** → positions chosen using random numbers so the choice is not biased **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	
	Model answer - Level 3 (8 marks)	
	A valid investigation begins with a clear, testable hypothesis and a null hypothesis, informed by existing knowledge, so that the data collected can actually accept or reject it. The first design decision is sample location: random sampling (grid coordinates or numbered sites) or systematic sampling along transects avoids the bias that would come from choosing convenient sites, and a preliminary study shows the interval needed to detect variation without collecting unnecessary data.	
	Timing matters because many variables change diurnally, seasonally or with weather, so samples on different occasions are needed and the interval must match the rate of change. Sample size and number depend on how variable the population is: a running mean shows when enough samples have been taken and the standard deviation shows the scatter around the mean. Techniques must be standardised so results from different sites, times or researchers are comparable, other variables controlled or monitored, and equipment calibrated so that precise readings are also accurate; anomalies should be repeated rather than ignored.	
	Finally the analysis must be chosen to suit the data - Spearman's rank for correlation, chi-squared for frequencies, a t-test for two means or Mann-Whitney U for medians - so that significance ($p < 0.05$) can be assessed, remembering that a significant correlation does not prove causation. Every decision trades reliability against cost, time and disturbance; sub-samples can only estimate the whole, so the most valid study is the one whose sampling design makes its limitations explicit.	
	Weaker answer - Level 2 (5 marks)	
	When planning fieldwork you need a hypothesis. You should choose sites randomly so it is not biased, for example using random numbers. You need to take enough samples and repeat them so the results are reliable. The method should be the same each time. You should measure accurately with calibrated equipment. Then you do a statistical test such as a t-test to see if the results are significant. If p is less than 0.05 the result is significant. This makes the conclusions valid.	
	What makes the difference	
	• Level 2 is sound but generic: it says 'enough samples' where Level 3 explains how a running mean and standard deviation tell you how many.	
	• Level 3 covers timing and the choice between statistical tests; Level 2 mentions one test only.	
	• The evaluation is missing from Level 2 - there is no discussion of trade-offs or of correlation vs causation, so it cannot reach the top level however accurate it is.	

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.1, 3.7.2.3.	
11.1	**Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Historical data limitations – proxy data - ice cores, isotope ratios (O-18:O-16, C-14), Mg:Ca in calcite, tree rings, pollen, coral • **Satellite sensors:** visible light imagery, infrared for vegetation and temperature, radar microwaves for wind, wave height, sea level and oil, weather monitoring – passing through cloud • **Sensor carriers:** ROVs / drones, AUVs, balloons, aircraft, animals – data recorders for abiotic factors – GPS tracking of floats and sampling sites – Argo floats and ocean monitoring • Climate models and their assumptions – monitoring atmospheric composition (Mauna Loa), ozone (Dobson units), sea level and ice • Applications to hydrosphere, minerals (remote sensing surveys), soils, pollution monitoring • **Evaluation:** improved coverage and resolution vs cost, calibration, interpretation, remaining uncertainty – justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 9, 89-107, 402-407; AQA spec 3.2, 3.7.	

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Question	Marking guidance	Marks
11.2	**Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Hydrological cycle processes and reservoirs – residence times – dynamic equilibrium – increasing demand (population, affluence, irrigation, industry) • **Human impacts on transfer rates:** deforestation, agriculture, urbanisation, climate change, abstraction – consequences for rivers, aquifers (salt incursion, subsidence, wetland loss), reservoirs • **Case studies of unsustainable exploitation:** Aral Sea, Nile, Tibetan rivers, Ogallala, North China Plain, Malta • **Sustainable strategies:** metering, low-water appliances, greywater, xeriscaping, drip irrigation, leak repair – new sources: rainwater catchment, reservoirs and estuary barrages (site factors), unexploited aquifers, inter-basin transfers, artificial recharge, river regulation, afforestation, desalination • Water treatment to make more water usable: sedimentation, flocculation, filtration, activated carbon, sterilisation, ion exchange, reverse osmosis – pollution control (sewage, AMD, buffer strips)	25
	• Links to thermohaline circulation and climate – energy costs – circular economy principles (catchment management) • Evaluation of which strategies are most sustainable in different regions – conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-128; AQA spec 3.2.2.	

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